Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Department of Computer Science and Engineering (Data Science)

Mini Project Report

TravelSync

Collaborative Travel Itinerary Planner

React, Node.js, Express.js and MongoDB

Submitted by:

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Class / Division: _____

Academic Year: _____

Subject / Course: _____

Project Guide: _____

Submission Date: _____

GitHub Repository:

https://github.com/Prathzz11/travel-itinerary-planner

1. Problem Statement

Group travel planning often involves separate chat messages, spreadsheets, booking confirmations and payment notes. Information becomes scattered: members may follow different schedules, lose reservation details or disagree about shared expenses. Manual calculations become difficult when several travellers pay for different items or make partial repayments.

The problem is to develop a centralized web application that allows a group to create a trip, organize a day-wise itinerary, maintain booking records, collaborate through controlled access and track shared expenses. The application must present consistent member balances and clearly distinguish trip expenditure from repayments between members.

1.1 Proposed Solution

TravelSync provides a common workspace for each trip. The Control Room lists personal trips, Trip Plan organizes daily activities, Resource Allocation tracks budgets and expenses, Settle Up records repayments, Logistics stores reservations, and The Crew manages participants. The Atlas provides public itinerary templates that can be copied and adapted.

2. Objectives

1. Provide authenticated access to personal and shared travel plans.
2. Create and edit trips with destinations, dates, budgets, currency and optional cover photos.
3. Organize activities by day, time, duration, category and location.
4. Maintain booking details and role-based collaboration within each trip.
5. Record shared expenses using equal or custom splits and calculate reliable member balances.
6. Support full or partial repayment records with payment history, correction and undo.
7. Enable discovery and reuse of public itinerary templates, with import and export support.

3. Modules and Process Explanation

3.1 Authentication and Profile

Users register and sign in with an email address and password. Passwords are hashed using bcrypt. The server issues a JSON Web Token (JWT) for protected API requests. Profile and signed-in password updates are available.

3.2 Trip Management

The Control Room displays trips with search, sorting and upcoming/past filters. Trip creation collects destination, dates, budget, currency and visibility. A JPG, PNG or WebP cover photo up to 2 MB, or an image URL, can be supplied and changed later.

3.3 Trip Plan

Each trip has dated days containing activities. An activity records its name, time, duration, category, location, notes and completion status. Timeline and calendar views organize the schedule; map features depend on a configured Google Maps key.

3.4 Resource Allocation

Expenses record an amount, category, payer and participating members. Equal splits distribute rounding differences consistently; custom splits must match the expense total. The page shows total expenses, remaining budget, category totals and member balances.

3.5 Settle Up

The page presents personal and whole-group balances, suggested repayments, and separate payment and expense histories. Editors can add an expense or record a payment directly. Recording a repayment changes the participants' outstanding balances, while total trip expenditure remains unchanged.

3.6 Logistics and The Crew

Logistics stores booking and confirmation details for travel arrangements. The Crew maintains members and roles. Owners and trip administrators manage trip details and membership; editors update trip records; viewers have read access. Only the owner can delete the trip.

3.7 The Atlas and Data Portability

The Atlas lists public templates with destination, duration, photos and planning information. National (India), international and total-budget filters can be combined with search. Lowest-budget sorting compares estimates within a selected currency. Users can inspect and copy templates into their own trips. JSON/CSV import and itinerary export reduce repeated manual entry. Public templates do not expose private expense or booking records.

3.8 Architecture and Data Processing

Layer	Technology and responsibility
Presentation	React, Vite and Bootstrap: pages, forms, navigation and shared UI state.
Service	Axios API client: authenticated HTTP requests and error handling.
Application	Node.js and Express: routes, authentication, permissions and validation.
Persistence	MongoDB and Mongoose: document schemas and stored trip records.

3.9 Main Data Entities

User: account and profile. **Trip:** owner, members, dates, budget and payments. **Itinerary:** dated days and activities. **Expense:** payer and split shares. **Booking:** reservation record. **Template:** public numbered-day plan. **Review:** template feedback and helpful votes.

3.10 Request and Synchronization Process

A page submits a request through a React context and the API client. Express checks the JWT, trip access and input values. A controller reads or writes data using Mongoose and returns the result. The frontend updates its shared state so related views use the latest confirmed information. Active trip and dashboard pages also refresh every 15 seconds and on focus or reconnection; refresh pauses during editing, while hidden, or when offline.

3.11 Expense and Settlement Calculation

For each member, the balance is calculated as:
Net balance = expenses paid - assigned expense shares + repayments sent - repayments received.
A positive result means the member gets money back; a negative result means the member owes money. Calculations use integer cents to avoid floating-point rounding differences. Suggested transfers match members who owe with members who are owed.

Example: one member pays INR 10,400 for four people. Each share is INR 2,600. The payer is owed INR 7,800 and the other three members each owe INR 2,600. After one member repays INR 2,600, that member is settled and the payer is owed INR 5,200. Trip expenditure remains INR 10,400.

3.12 Validation and Scope

The API checks membership, amounts, split totals and currency. Payment request IDs make retries safe, and stale edits are rejected. Bookings and repayments are user-maintained records; the application does not purchase travel or transfer money. Optional cross-document transaction mode requires a MongoDB replica set.

4. Flow Diagram

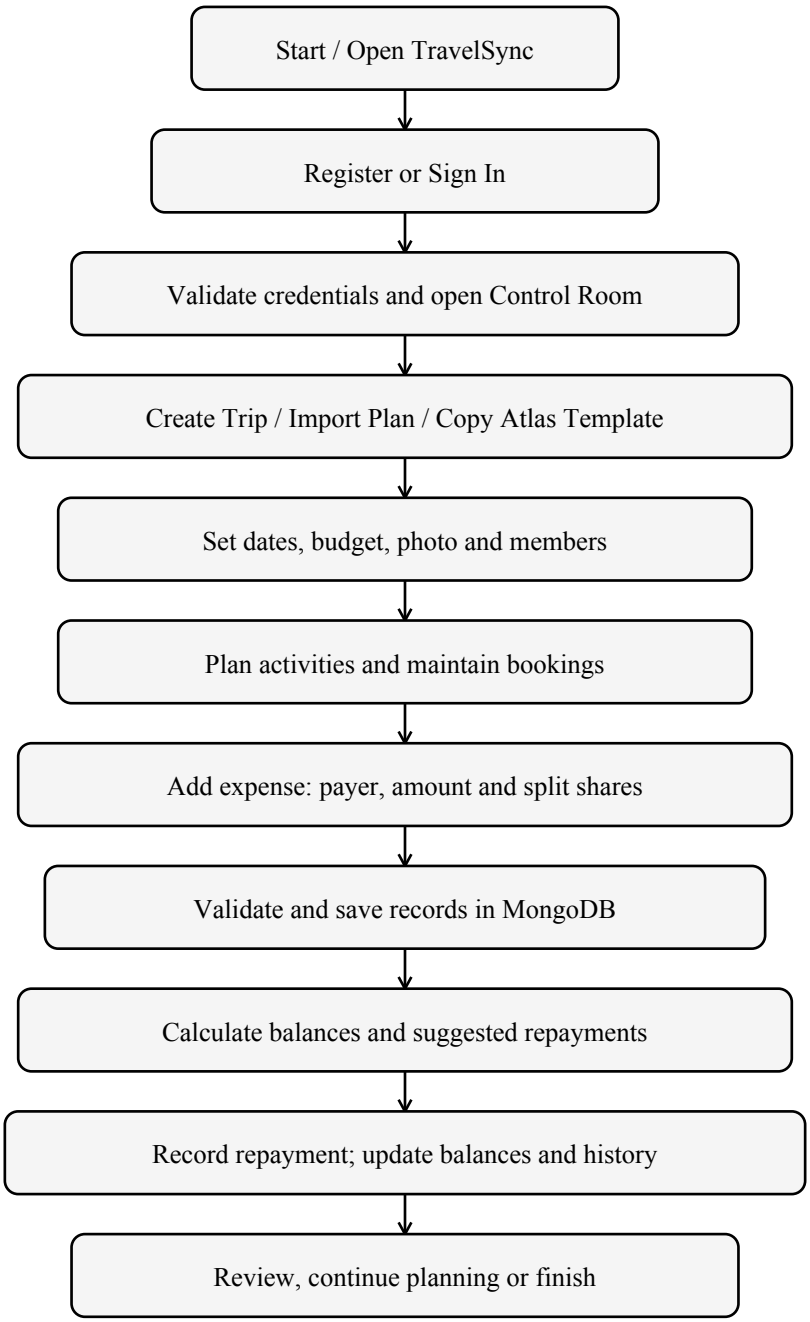


Figure 1. Main trip-planning and settlement workflow.

Invalid credentials or invalid form values return an error for correction before the next step. Members can revisit planning, expenses and payments throughout the trip.

5. Implementation (Screenshots)

The following figures show the running application with temporary demonstration data. The sample trip has four members. Screenshots were captured on 26 October 2026.

5.1 Authentication

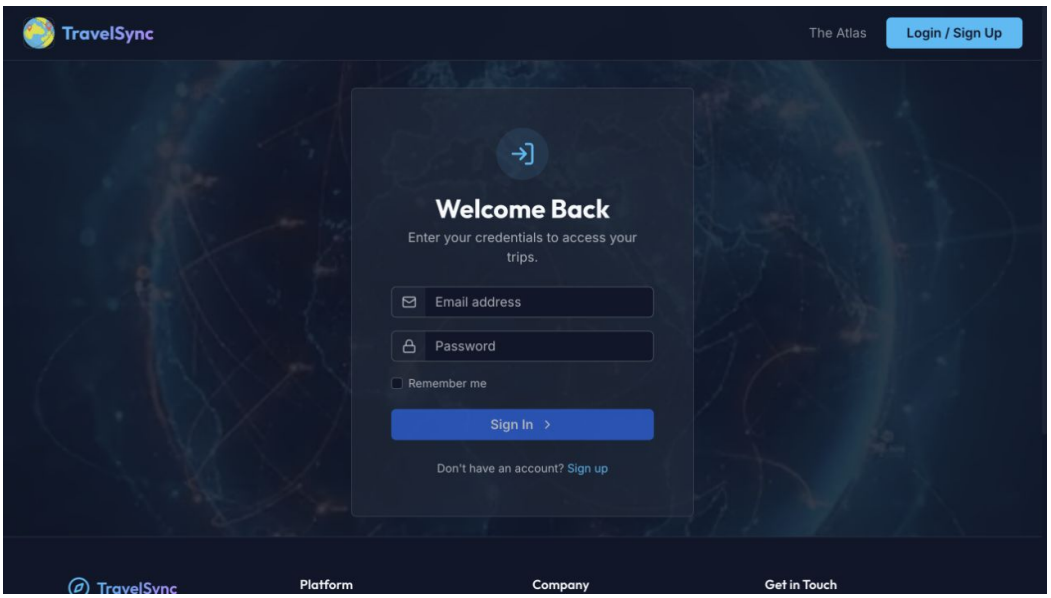


Figure 2. Login screen for authenticated access.

5.2 Creating a Trip

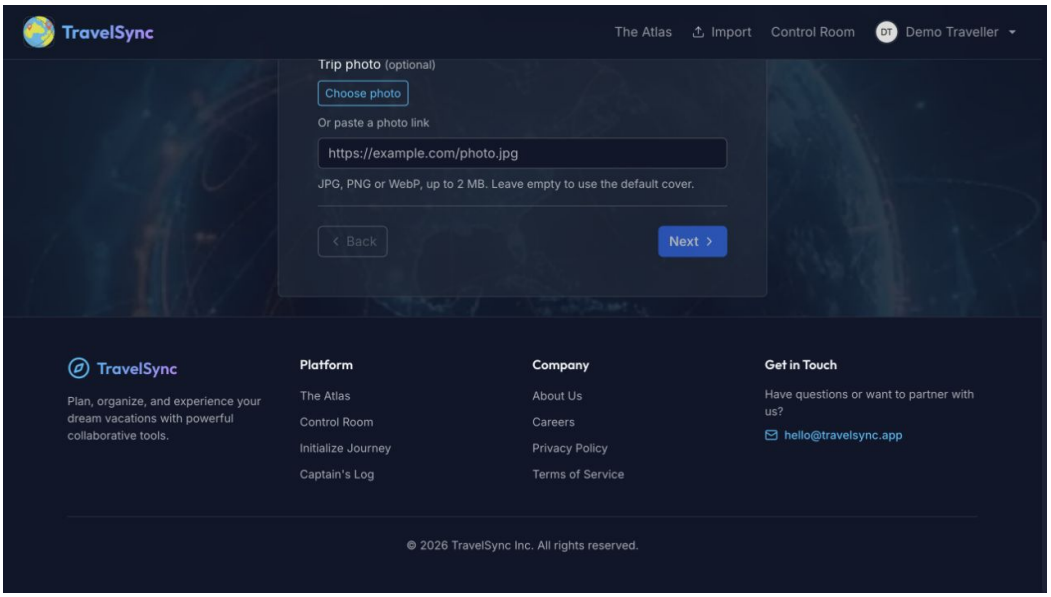


Figure 3. New-trip form with optional photo upload or image link.

The remaining creation steps collect trip dates, budget and settings. Cover photos can later be replaced or removed from the trip edit form.

5.3 Control Room and Trip Plan

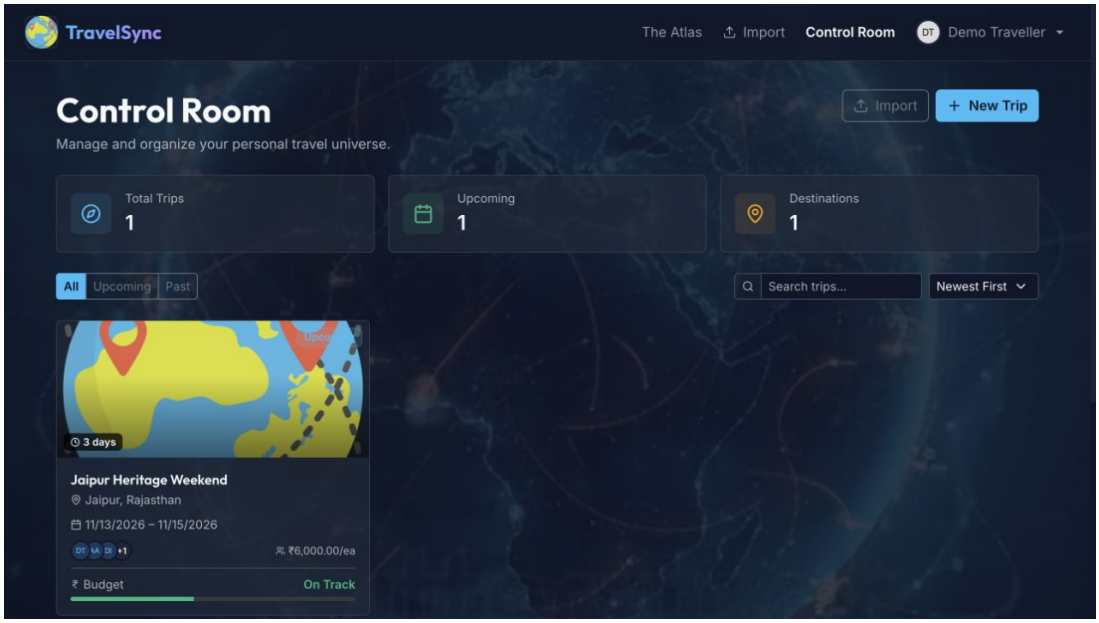


Figure 4. Control Room displaying the demonstration trip.

The dashboard provides a central entry point to existing trips and supports creating a new trip, searching, filtering and importing a plan.

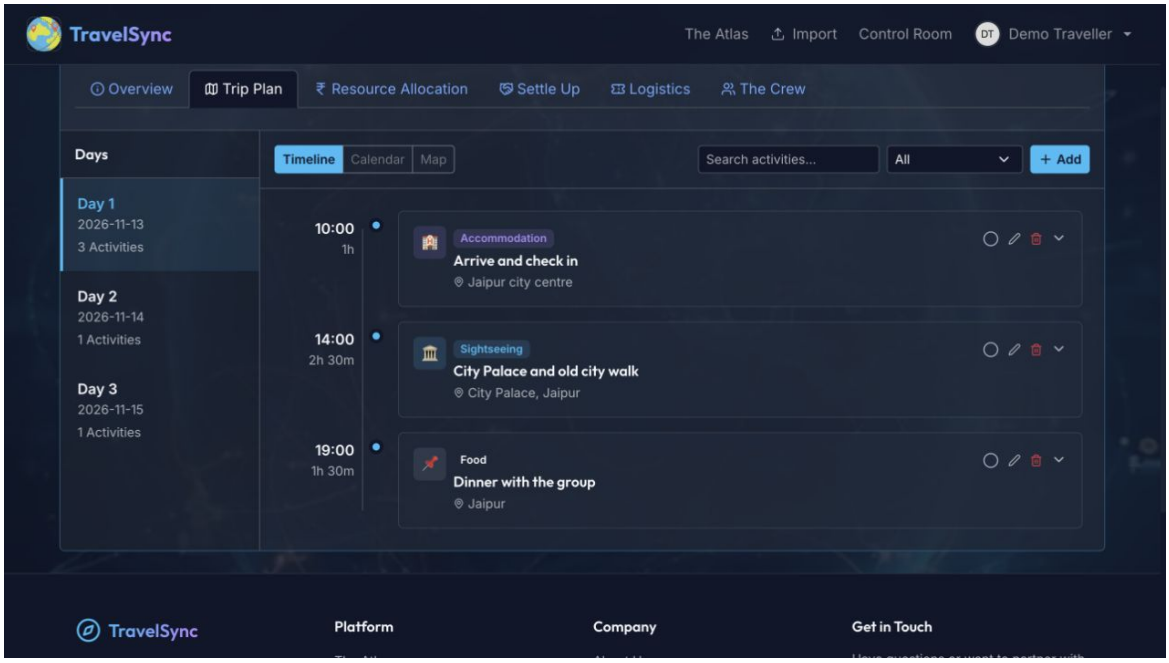


Figure 5. Day-wise timeline with activity times, locations and editing controls.

The Trip Plan screen groups activities by date. Users can add or modify activities and mark completion as the trip progresses.

5.4 Shared Expenses and Budget

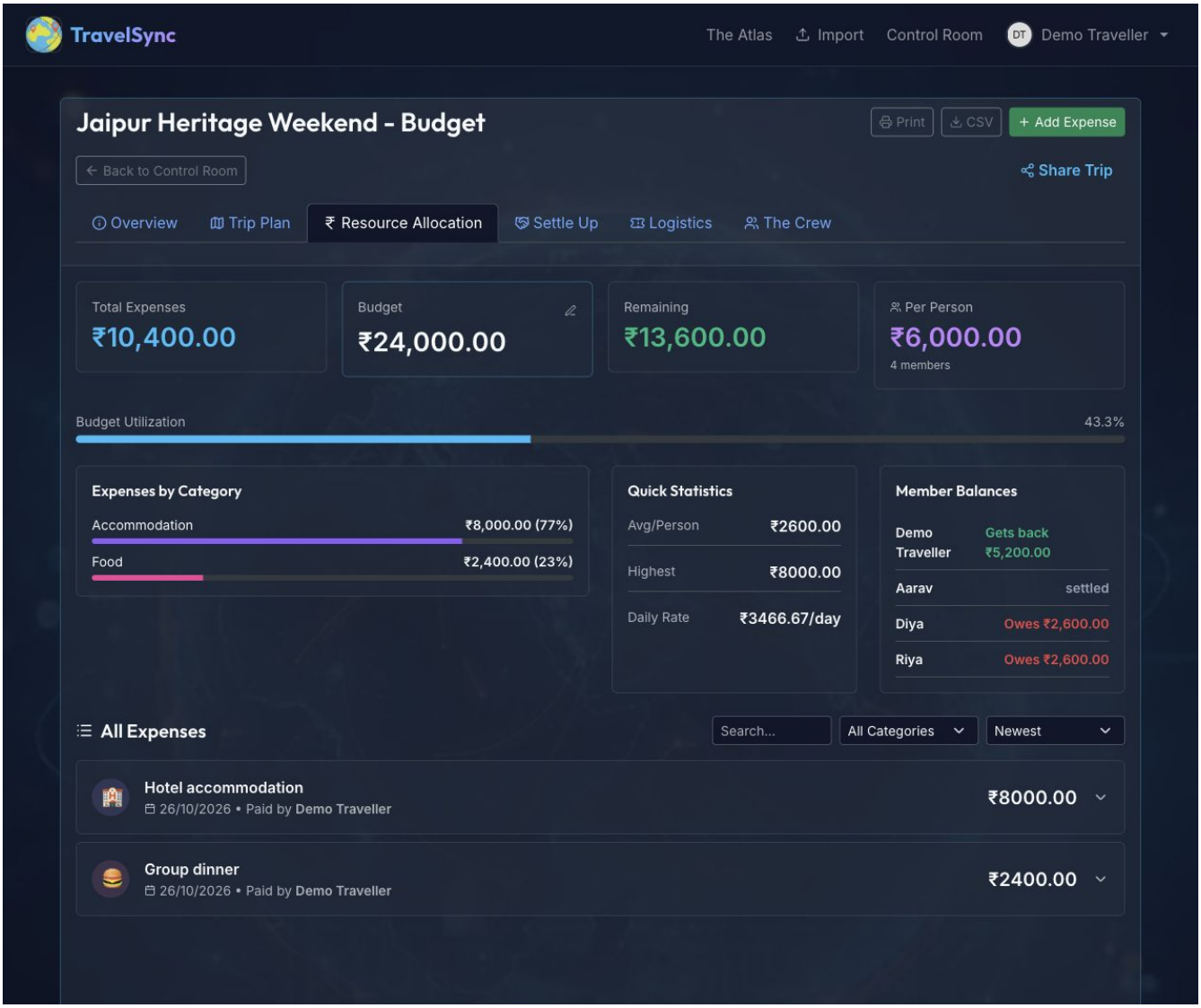


Figure 6. Expense breakdown and member balances after the sample repayment.

Hotel accommodation of INR 8,000 and a group dinner of INR 2,400 produce total expenses of INR 10,400. Against a budget of INR 24,000, the remaining amount is INR 13,600. The balance panel also reflects the repayment recorded in Settle Up.

5.5 Settle Up - Outstanding Balances

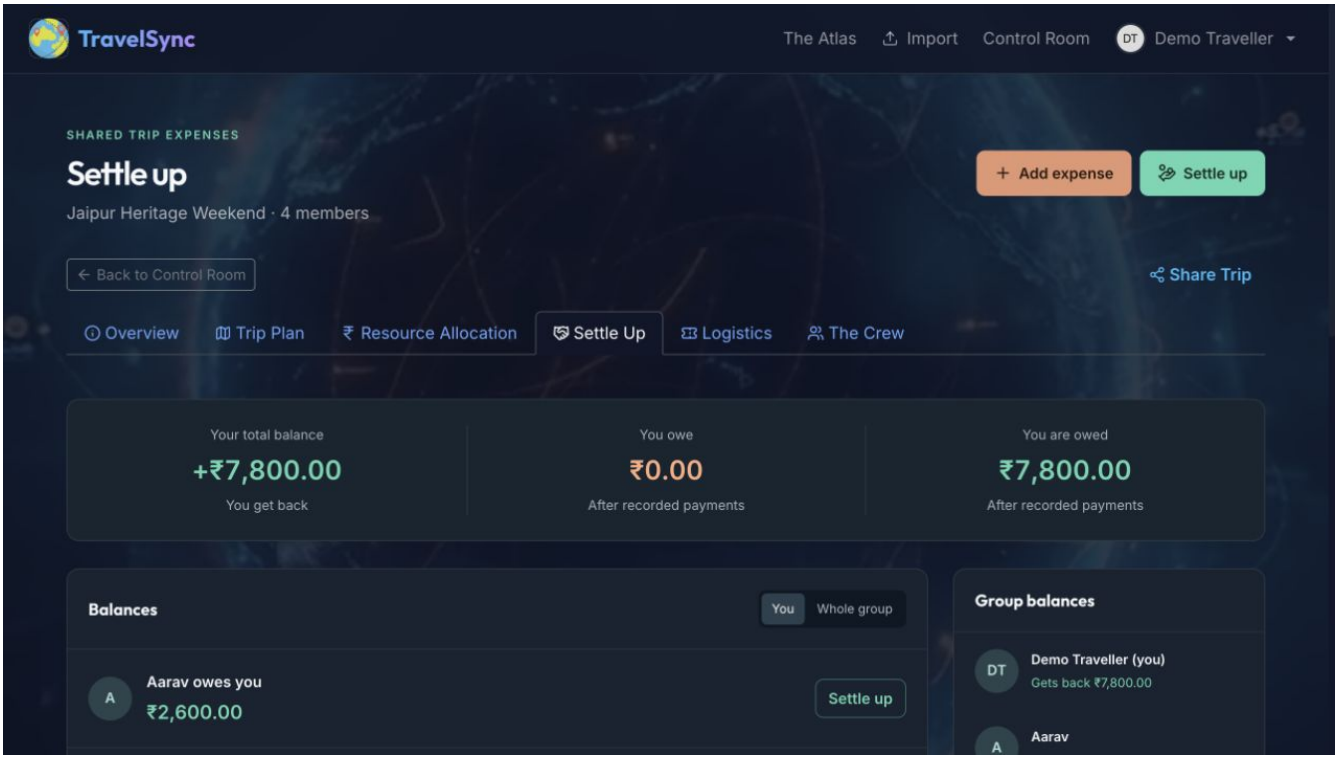


Figure 7. Personal balance summary with Add expense and Settle up actions.

How the Page Is Used

1. Open Settle Up to see the amount owed or receivable.

2. Select a suggested member payment or use the general Settle up action.

3. Confirm the payer, recipient and amount; a smaller amount records a partial payment.

4. Save the payment and review the updated history and balances.

Before the demonstration payment, Demo Traveller is owed INR 7,800. Each of the other three members owes INR 2,600. The page separates repayment history from expense records to keep spending and settlement calculations clear.

5.6 Recording a Repayment

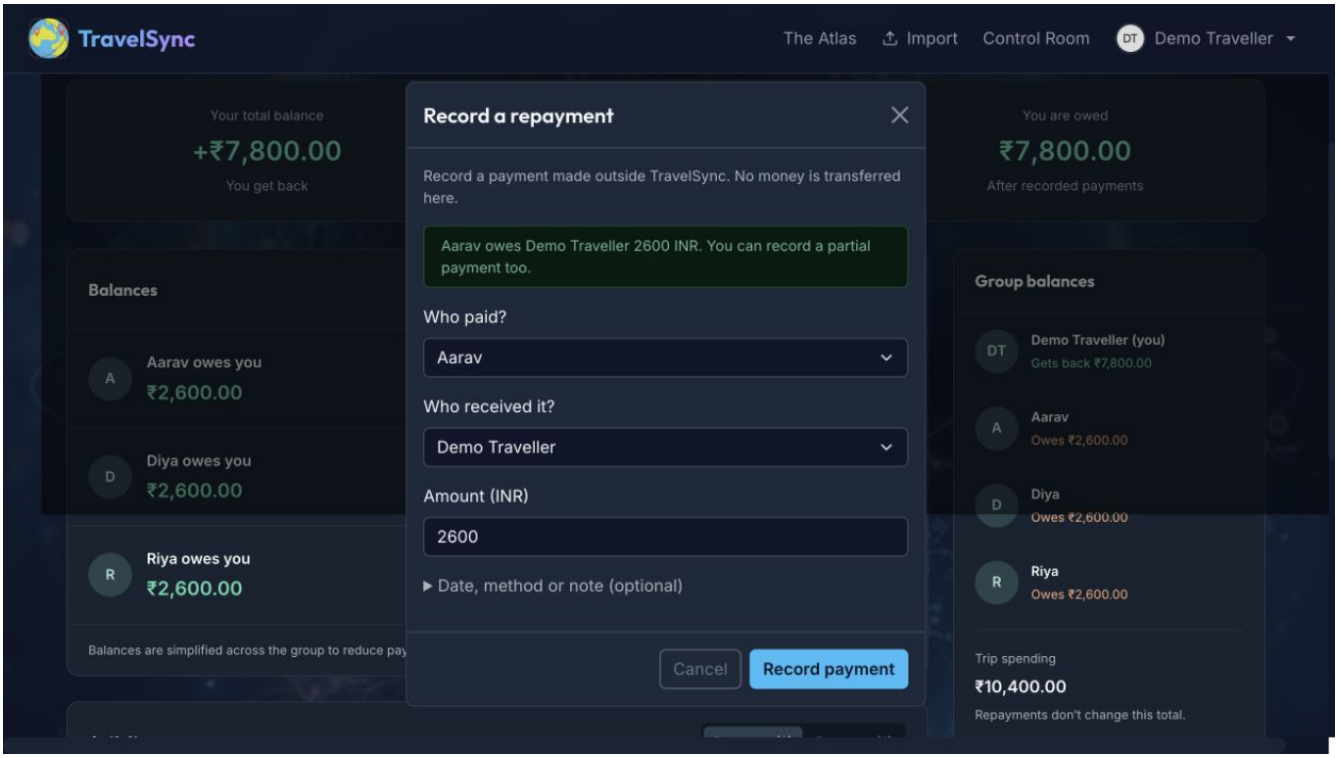


Figure 8. Prefilled repayment form for Aarav paying Demo Traveller.

The form uses the suggested payer and recipient and pre-fills the outstanding amount of INR 2,600. The amount remains editable for partial repayments. Date, payment method and notes are optional supporting information.

Selecting Record payment saves a record of a payment made outside TravelSync. No bank transaction or money transfer occurs through this screen.

Correction and Consistency

Recorded payments can be edited when details are incorrect or undone when recorded by mistake. The backend validates member identities and amounts. Version checks prevent an outdated edit from silently replacing a newer record.

5.7 Result and Verification

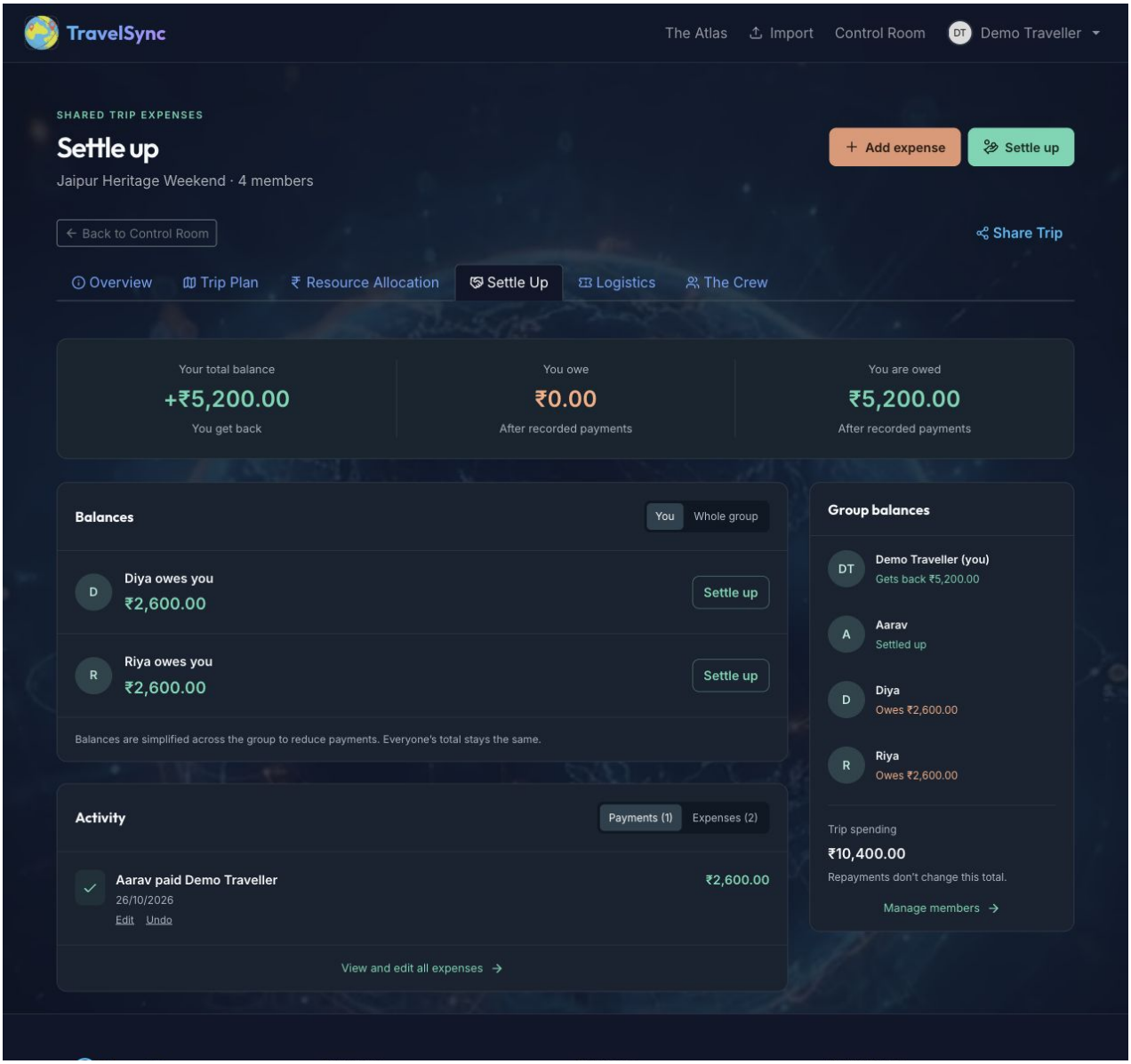


Figure 9. Recorded payment, updated group balances and unchanged trip spending.

Observed result: Aarav is marked settled; Demo Traveller is owed INR 5,200; Diya and Riya each owe INR 2,600. Payment history contains the INR 2,600 record, and trip spending remains INR 10,400. The Resource Allocation screen shows the same member balances (Figure 6).

These observations verify the illustrated demonstration flow. They do not imply that every possible input, concurrent edit or external service has been exhaustively tested.

5.8 The Atlas and Project Outcome

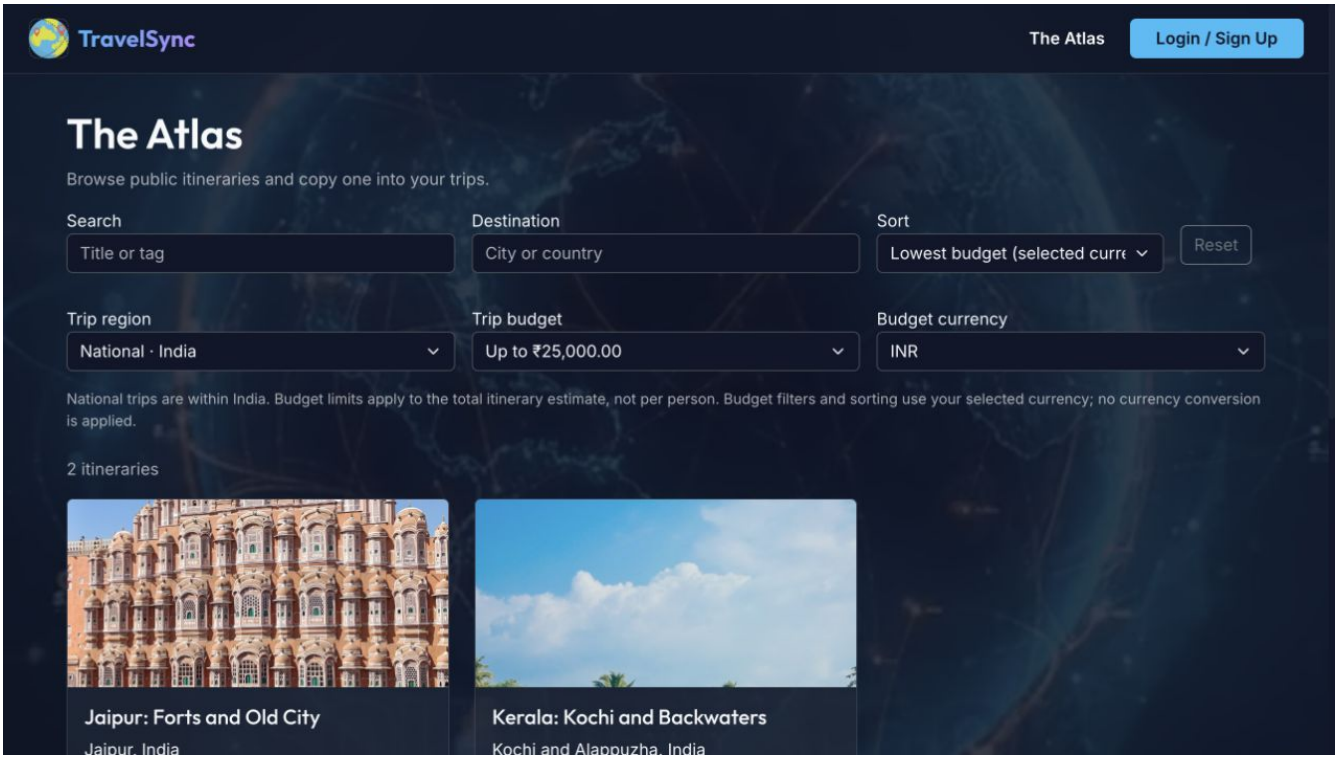


Figure 10. National trips under INR 25,000, sorted by lowest budget.

The Atlas now combines destination search with National (India), International, total-budget and currency filters. Lowest-budget sorting uses the selected currency without conversion. The illustrated results show Jaipur at INR 18,000 and Kerala at INR 22,000. Filters are applied by the API before pagination; Reset restores all results. Unrecognized destinations remain under All destinations.

5.9 Outcome and Future Scope

TravelSync brings trip planning, shared expenditure and repayment records into one application. The implementation provides a clear sequence from trip creation to itinerary preparation, expense tracking and settlement. Shared state and periodic refresh help related screens remain consistent.

Future improvements may include password recovery, invitation-email delivery, push-based collaboration and a dedicated image-storage service. Maps require a valid provider configuration. Production deployments should use HTTPS, protected configuration and an appropriate MongoDB setup.

Implementation Basis

Source code: github.com/Prathzz11/travel-itinerary-planner (branch: backend). This report is based on the project source and live demonstration screens. The active API is in backend/; the server/ directory is a legacy copy. UI screenshots show demonstration records rather than personal financial data.